

Algebraic Vision: A Vision Transformer Without Transcendental Functions That Outperforms the Standard ViT

Tasmai Keni
tas.ken.rt25@dypatil.edu

Abstract

We build a Vision Transformer whose forward pass, backward pass and training loss use only arithmetic, square roots and comparisons, and compare it with a standard ViT under matched data, parameters, token budget, optimizer and tuning effort. Softmax attention is replaced by a radical kernel $E_n(s) = (s/n + \sqrt{1 + s^2/n^2})^n$ that agrees with e^s to second order but grows polynomially, so attention weights keep a provable, scale-independent floor; the GELU gate by a radical logistic; learned absolute position embeddings are complemented by learned two-dimensional Cayley rotations; and cross-entropy by a strictly proper power score. The compiled training graph contains no exponential, logarithm or other transcendental instruction. Properties used by the design are checked symbolically and in Lean 4. In short-budget ImageNet-1k training the algebraic model trains stably at learning rates where the standard ViT degrades or diverges. At the vision counterpart of a 125M-parameter, 2.5B-token language-model study (ViT-B/16, 224px, 2,500,169,728 patch tokens, three seeds) it reaches 68.19% validation top-1 against 58.32% for the standard ViT (paired difference +9.86 points, one-sided paired t -test $p = 2.2 \times 10^{-4}$) and 65.87% for a standard ViT with two-dimensional rotary positions, with each model's learning rate selected on held-out training images and the standard model searched on a wider grid.

1 Introduction

Transformers combine matrix products with a few nonlinear operators that are usually taken as fixed: the exponential inside softmax attention, a sigmoid-family or Gaussian gate in the MLP, sines and cosines in rotary positions, and the logarithm of cross-entropy [10, 12, 20, 23]. A preceding study [14] showed that a causal language model built only from algebraic operations can be trained at 125M parameters and reach perplexity close to a matched standard Transformer. This paper asks a stronger question for vision: *can an algebraic design be more accurate than the standard one*, under a protocol that does not favour it?

Three properties make the algebraic setting more than a constraint. Radical expressions are cheap on accelerators (one inverse square root and a few multiplications); they come with polynomial rather than exponential tails, which changes how attention behaves as scores grow; and their real-arithmetic identities are short enough to be machine-checked. Our contributions are:

- **Radical attention** (section 4.1): weights $E_n(s_{ij})/\sum_k E_n(s_{ik})$. We prove that for positive scores the ratio of two weights is at most $(s_1/s_2)^n$ at every score scale and that every weight keeps a floor $(a/b)^n/N$ when scores lie in $[a, b]$; softmax has neither. As scores grow, softmax rows saturate and their gradients vanish while radical rows do not (fig. 2).
- A complete algebraic ViT (section 4): a radical-logistic gate matching GELU, learned 2-D Cayley rotations giving exactly relative position logits, and a strictly proper power-score loss with an overflow-proof radical link.
- A fair, preregistered comparison (section 5): identical data order, augmentation, initialisation stream, optimizer and budget; at least as much tuning for the standard model; selection on a held-out split of the training set; paired seeds; a compiled-graph purity audit; and a relative-position control.
- Results from small trials to ViT-B/16 (section 6): the algebraic model wins at every stage, by +7.9 points in small-scale trials, +11.0 points in the pilot and +9.86 points in the main comparison. Relative positions account for most of the main-run gain: a standard ViT with 2-D rotary positions closes much of the gap, and the algebraic model leads it by 2.31 points.

2 Related work

Alternatives to softmax attention. Sparsemax and α -entmax [17, 18] replace the exponential by thresholded polynomials and produce sparse weights; pointwise attention without normalisation, such as ReLU attention divided by the sequence length [24], removes the exponential altogether. Radical attention keeps a dense, normalised and softmax-like distribution, which agrees with softmax to second order for small scores, and changes only how fast the kernel grows.

Instability of attention logits. Large learning rates and long training let attention logits grow until softmax heads collapse to one-hot weights, which stalls training; the remedies include controlling attention entropy [25] and normalising queries and keys [9]. Dot-product attention is also not Lipschitz [15]. The polynomial tail of E_n addresses the same failure through the kernel itself (proposition 1). Attention sinks and register tokens [7] give heads a place to put mass they do not need; a learned sink did not help in our budgets.

Positions, losses and baselines. Rotary embeddings [20] and their two-dimensional variants for ViTs [13] make logits depend on relative positions through sines and cosines; Cayley rotations give the same structure with rational functions of a learned parameter. Cross-entropy is the logarithmic member of the strictly proper scoring rules [11], and Fenchel–Young losses [3] generalise it to other regularisers; we use the power family because its fractional powers reduce to square roots. The standard model is a plain ViT in the style of [2, 10]; neither model uses the stronger augmentation and regularisation of [22]. All code uses JAX [4].

3 The algebraic constraint

The algebraic model’s forward pass, backward pass and loss use only $+$, $-$, $\times$, $\div$, square roots (through `rsqrt`), $\min$, $\max$, $|\cdot|$, comparisons and selects. We check this on the program the TPU actually runs: the model, loss and gradient of ViT-B/16 are compiled for TPU v4 and every HLO instruction is scanned. The algebraic graph contains no exponential, logarithm, logistic, tanh, error function, trigonometric function or real power (208 `rsqrt` instructions); the standard graph contains 40 exponentials and a logarithm. Two parts are outside the constraint and shared by both models: the optimizer (AdamW with a cosine schedule [16]) and the image pipeline. Top-1 accuracy is an argmax; the logarithm appears only when we report negative log-likelihood.

4 Algebraic components

4.1 Radical exponential and attention

For $n = 2^m$ let

$$\rho_n(s) = \frac{s}{n} + \sqrt{1 + \frac{s^2}{n^2}}, \quad E_n(s) = \rho_n(s)^n. \quad (1)$$

E_n costs one inverse square root and m squarings. It is positive, $E_n(s)E_n(-s) = 1$, and $E_n(s) = 1 + s + \frac{s^2}{2} + (1 - n^{-2})\frac{s^3}{6} + O(s^4)$, so it agrees with e^s to second order and $E_n \rightarrow e^s$ as $n \rightarrow \infty$; its logarithmic derivative is $g_n(s) = (1 + s^2/n^2)^{-1/2} \in (0, 1]$. For $s \gg n$, however, $E_n(s) \sim (2s/n)^n$, and $E_n(-s) = 1/E_n(s)$. Attention uses

$$p_{ij} = \frac{E_n(\beta s_{ij})}{\sum_k E_n(\beta s_{ik})}, \quad s_{ij} = \frac{\langle q_i, k_j \rangle}{\sqrt{d_h}}, \quad (2)$$

with $n = 8$ and $\beta = 1$ in the final model.

Proposition 1 (scale-free ratios and an attention floor). *For $0 < y \leq x$ and every $c > 0$, $E_n(cx)/E_n(cy) \leq (x/y)^n$ (`kernel_ratio_scale_free`), whereas e^{cx}/e^{cy} is unbounded in c (`exp_ratio_unbounded`). If all N scores of a row lie in $[a, b]$ with $a > 0$, then for every scale $c > 0$ each weight satisfies $p_j \geq (a/b)^n/N$ (`attention_floor`).*

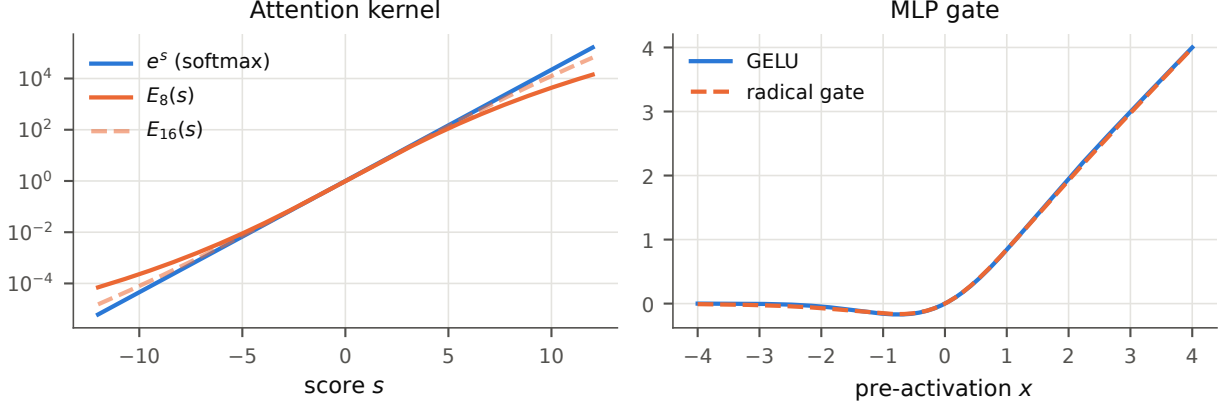


Figure 1: Left: E_n against e^s (log scale); they agree near zero and E_n grows polynomially for large $|s|$. Right: the radical-logistic gate against GELU.

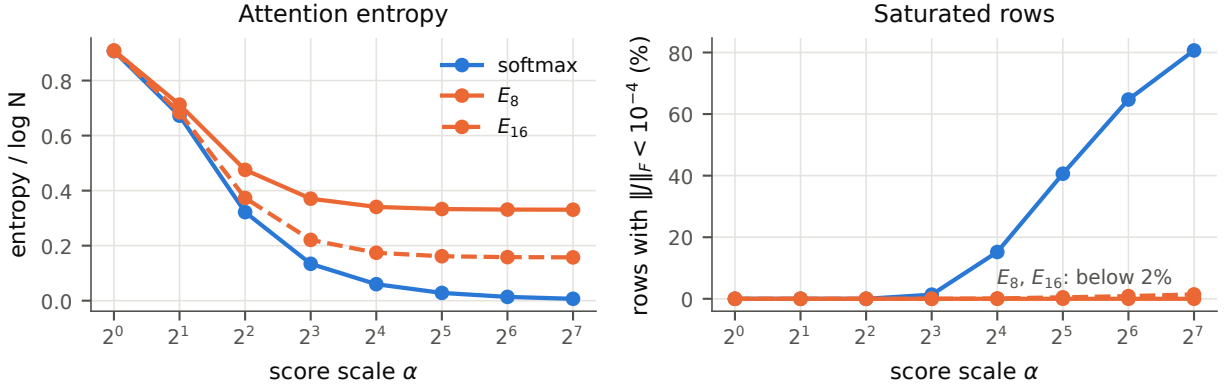


Figure 2: Attention on 10^4 random rows of 197 Gaussian scores multiplied by α . Softmax entropy collapses and most rows saturate; radical attention keeps a floor and a non-vanishing gradient.

The proof reduces to $y\rho(x) \leq x\rho(y)$, i.e. $y^2(n^2+x^2) \leq x^2(n^2+y^2)$, plus monotonicity of ρ . The consequence is that growing query–key norms cannot drive radical attention to a one-hot distribution: sharpening requires multiplicative score gaps. Softmax attention can collapse as logits grow, a known instability of Transformers trained with large learning rates [9, 25]. The score Jacobian is $\partial p_i / \partial s_j = \beta g_n(\beta s_j) p_i(\delta_{ij} - p_j)$, with entries at most $\beta g_n/4$ (`jacobian_diag_le`, `jacobian_offdiag_le`). Under score scaling (fig. 2) softmax entropy falls to 0.007 of its maximum and 81% of rows become saturated (Jacobian norm below 10^{-4}), while E_8 keeps a normalised entropy of 0.33 and 0% saturated rows.

Two further properties matter for implementation. Numerators and denominators accumulate over key tiles exactly, without the running maximum and rescaling of tiled softmax [6] (`tilled_eq_dense`); E_8 overflows float32 only for $|s| > 2.6 \times 10^5$. And radical attention is better conditioned: rounding its inputs to bfloat16 perturbs the weights up to an order of magnitude less than it perturbs softmax weights at large scores (table 3). We tried a per-head “sink” term in the denominator, which lets a head attend to nothing; it was measurably worse in our short budgets and is not used.

4.2 MLP gate

With $\sigma_n(y) = 1/(1 + E_n(-y))$ we use $\text{rgelu}(x) = x \sigma_8(1.702x)$, the algebraic counterpart of the logistic approximation of GELU. $\sigma_n(y) + \sigma_n(-y) = 1$, σ_8 differs from the logistic function by $-y^3/1536 + O(y^5)$

at the origin, and $\sigma'_n = \sigma_n(1 - \sigma_n)g_n \leq 1/4$ (`radicalSigmoid_symm`, `sigmoid_slope_1e`). An earlier gate, $x(1 + cx/\sqrt{1 + c^2x^2})/2$ with $c = \sqrt{2/\pi}$, has a smaller Lipschitz constant (1.044 against GELU’s 1.129, `abs_dalu_1e`) but polynomial tails that make it too soft; it cost 1.6 points (table 1) and was replaced.

4.3 Two-dimensional Cayley rotations

The Cayley matrix $R(w) = \frac{1}{1+w^2} \begin{pmatrix} 1-w^2 & -2w \\ 2w & 1-w^2 \end{pmatrix}$ is a rotation by $2 \arctan w$ (`cayley_orthogonal`, `cayley_det`). Each feature pair $(x_i, x_{i+d_h/2})$ of every head is turned by $R(w_i^x)^c R(w_i^y)^r$ at grid cell (r, c) , with learned w^x, w^y per head and pair, applied to queries and keys. Powers are formed by repeated multiplication with renormalisation. Because planar rotations commute, attention logits depend only on the two-dimensional offset between query and key (`mixed_cayley_relative`). The model keeps learned absolute position embeddings as well. The learned frequencies add 9,216 parameters to ViT-B/16 (0.01%).

4.4 Loss

Predictions use the radical link $p_c = E_{64}(z_c)/\sum_k E_{64}(z_k)$, evaluated through ratios to the row maximum of ρ , which cannot overflow. The loss is the Bregman divergence of $\phi(t) = (t^a - t)/(a(a - 1))$:

$$L_a(p, y) = \frac{1}{a} \sum_c p_c^a - \frac{1}{a-1} \sum_c y_c p_c^{a-1} + \text{const}(y), \quad a = \frac{63}{64}, \quad (3)$$

with label-smoothed targets y . It tends to the log score as $a \rightarrow 1$ and is strictly proper for every $a > 0$, $a \neq 1$ [11]; the Lean proof (`power_score_strictly_proper`) rests on Bernoulli’s inequality. The powers $p^{2^{-6}}$ are six nested square roots of the partition sum. With $a = 7/8$ and an E_8 link this loss cost two points against cross-entropy; with $a = 63/64$ and E_{64} it matches cross-entropy. Fenchel–Young losses [3] with algebraic links (squareplus, 1.5-entmax [18]) failed at 1000 classes because of heavy or sparse tails.

5 Protocol

Models. Both models are pre-norm ViTs with LayerNorm [1], global average pooling, a zero-initialised linear head, and the same patch embedding, widths, depth and heads. The standard model is the canonical ViT [2, 10]: softmax attention, a GELU MLP, learned absolute position embeddings and cross-entropy with label smoothing 0.1. The algebraic model swaps in the components above. Shared tensors start from identical values for a given seed.

Data and training. ImageNet-1k [19]; 10 images per class are held out from the training split as *minival* for every selection decision, and the 50,000-image validation set is used only for reporting. Both models see the same images in the same order with the same random-resized crops and flips for a given seed. AdamW ($\beta_2 = 0.999$, weight decay 0.05), cosine schedule with linear warmup, gradient clipping at norm 1, batch 1024 (512 for the small trials and the pilot), bfloat16 matrix products, TPU v4.

Stages. (1) Small-scale trials: ViT-S/8 on 64×64 ImageNet for 4 epochs [5], with every component swapped alone and together, paired seeds. (2) Pilot: ViT-S/4 (22M parameters, 256 tokens) for 10 epochs, 3.25B patch tokens. (3) Sweep: ViT-B/16 at 224 px, 600M patch tokens per run, peak learning rate $\{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ for both models, seeds 42–44, selection by mean minival. (4) Main runs: ViT-B/16, 2.5B patch tokens (≈ 10 epochs), the selected learning rates, seeds 42–44. The sweep and main-run specifications were committed before their first run.

Fairness. The standard model received at least as much tuning as the algebraic one at every stage: in the small trials nine learning-rate and warmup settings each; in the sweep the standard grid was extended by $\{2.5, 3.5, 7\} \times 10^{-4}$ after its best value sat at the lower edge (the algebraic grid was not extended, although its best value sat at the upper edge). A script checks that paired runs differ only in the arm-defining fields and the selected learning rate. Both models prefer a long warmup (40% of training) in this short-budget regime, and both use it.

Table 1: Component screen, ViT-S/8 on 64 px ImageNet, 4 epochs, learning rate 10^{-3} , validation top-1 (%), 4 seeds, Δ the mean paired difference. This learning rate is above the standard ViT’s best (5×10^{-4}), so the gain of radical attention here includes its robustness to the learning rate; best-against-best comparisons are in the text.

Configuration	Top-1	Δ vs. standard
Standard ViT	20.71	
+ radical attention	29.61	+8.89
+ 2-D Cayley rotations	25.27	+4.55
+ first algebraic gate (ALU)	19.14	−1.57
+ power loss $a = 7/8$, E_8 link	18.70	−2.02
Radical attention + learned Cayley (2 seeds)	34.37	

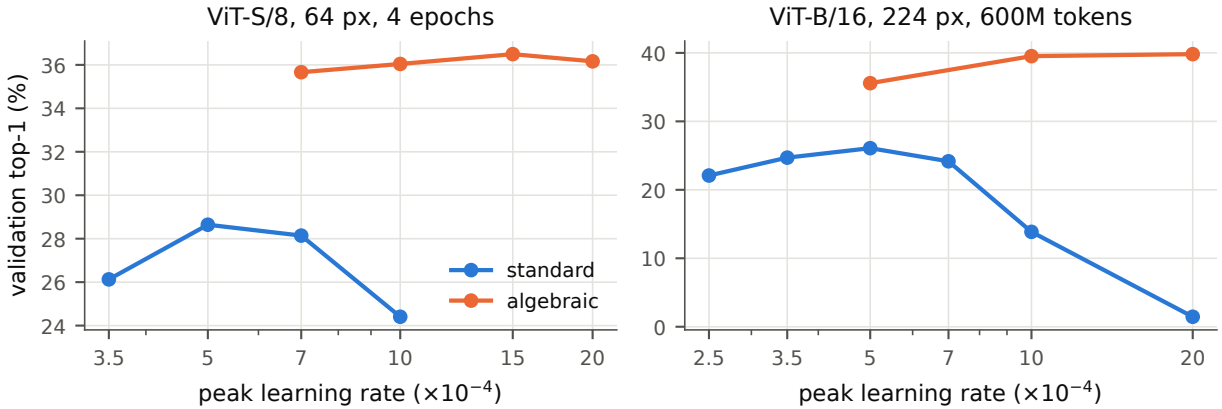


Figure 3: Validation top-1 against peak learning rate (mean over seeds). Left: ViT-S/8, 64 px, 4 epochs, 25% warmup, two to four seeds per point. Right: ViT-B/16, 224 px, 600M patch tokens, 40% warmup, three seeds per point; the standard grid was extended below and between its best values.

6 Results

Small-scale trials. Swapping single components into the standard ViT at learning rate 10^{-3} (table 1) identified radical attention and Cayley rotations as the gains and the first gate and loss as losses; the latter were redesigned until they matched GELU and cross-entropy in the stable regime. Over a learning-rate grid (fig. 3, left) softmax attention degrades above 5×10^{-4} (at 2×10^{-3} with a 10% warmup it collapses to 4.9%, where radical attention reaches 26.7%), while the algebraic model varies by less than a point from 7×10^{-4} to 2×10^{-3} . With 25% warmup and each model at its best learning rate, the algebraic model leads by +7.9 points on four paired seeds ($p = 2.7 \times 10^{-5}$): 36.49% against 28.64%. Adding sine/cosine 2-D rotary positions [13] to the standard ViT gives 32.58%, in between. A 40% warmup helps both models (29.15% and 37.60%, two seeds each), and is used from here on.

Pilot. At 10 epochs with ViT-S/4 (fig. 4) the algebraic model reaches 59.93% validation top-1 against 48.88% (+11.0 points on both seeds), leading at every checkpoint, with lower NLL (1.80 against 2.36).

Sweep. At ViT-B/16 the standard ViT is best at 5×10^{-4} (26.08% after 600M tokens), with a clear optimum: it loses one to two points at 3.5 and 7×10^{-4} , half its accuracy at 10^{-3} , and collapses at 2×10^{-3} . The algebraic ViT is best at 2×10^{-3} (39.82%) and within half a point at 10^{-3} (fig. 3, right). These are the learning rates of the main runs.

Table 2: Main comparison: ViT-B/16 at 224 px, 2,500,169,728 patch tokens per run. Validation top-1 (%) on 50,000 images, NLL and expected calibration error (ECE, 15 bins); mean $\pm$ standard error over seeds 42–44; throughput on 16 TPU v4 chips.

Model	lr	Top-1	NLL	ECE	img/s
Standard ViT	5×10^{-4}	58.32 ± 0.02	1.871	0.072	11,895
Algebraic ViT	2×10^{-3}	68.19 ± 0.23	1.410	0.068	10,311
Standard + RoPE-2D (control)	5×10^{-4}	65.87 ± 0.32	1.510	0.079	12,292

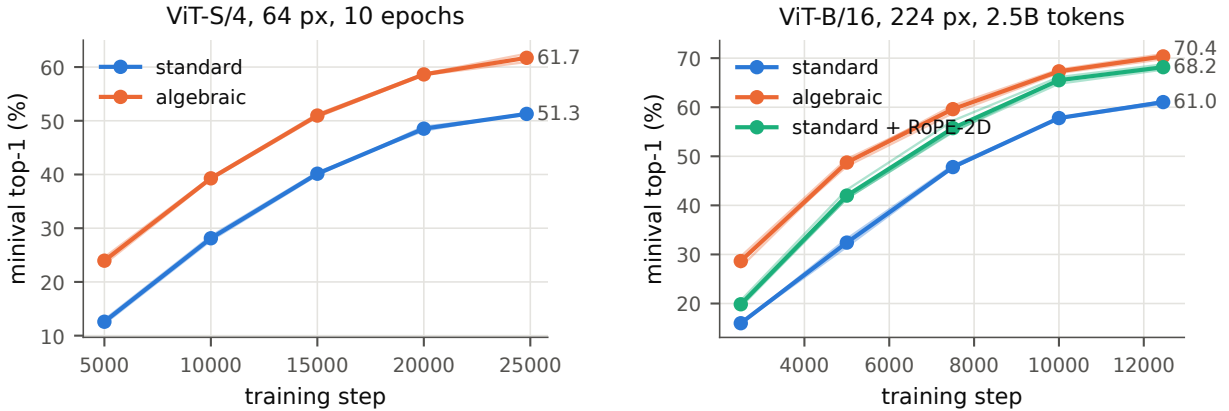


Figure 4: Minival top-1 during training: mean over seeds, thin lines are single seeds, numbers are final means. Left: pilot, ViT-S/4 at 64 px. Right: main runs, ViT-B/16 at 224 px, with the RoPE-2D control.

Main comparison. Table 2 and fig. 4 report the 2.5B-token runs. The algebraic model is ahead on all three seeds (by +9.96, +9.47, +10.16 points), by +9.86 points on average (95% interval $[+8.97, +10.75]$, one-sided paired t -test $p = 2.2 \times 10^{-4}$), with lower NLL and similar calibration. This meets the criterion fixed before the runs: every seed in favour, a positive mean difference and $p < 0.05$. It leads at every evaluation during training (fig. 4). The standard ViT with sine/cosine 2-D rotary positions, trained with the standard model’s hyperparameters, reaches 65.87% (+7.55 points relative to the standard ViT), 2.31 points below the algebraic model. Relative positions therefore account for most of the gain over the standard ViT. The algebraic model, whose Cayley rotations are the rational counterpart of these rotary positions, still leads the control on 3/3 seeds (95% interval $[+1.25, +3.37]$, $p = 0.006$); the control used the standard model’s learning rate and was not tuned separately.

Mechanism. In the trained models (fig. 5) the algebraic model’s attention scores are an order of magnitude larger than the standard model’s (median $|s|$ per block up to 94 against 11), yet its attention is broader in 9 of 12 blocks: in its middle blocks at most 0.6% of rows are near one-hot (largest weight above 0.9), against up to 7.9% for softmax. At such scores softmax would be one-hot; the radical kernel responds to score ratios rather than differences (proposition 1), so the model can use large scores without saturating, which matches its tolerance of large learning rates.

Cost. With the attention computed by XLA, the algebraic model trains at $0.87\times$ of the standard model’s throughput. Fused Pallas kernels for both attentions (one image per program, all heads in VMEM, analytic backward) run a ViT-B/16 attention layer forward and backward in 0.891 ms (softmax) and 1.132 ms (radical) on one TPU v4 chip, against 1.035 ms and 1.887 ms for XLA; the remaining gap is the radical kernel’s extra multiplications per score. The training runs use the XLA attention for both models, and all comparisons are at equal tokens, not equal time.

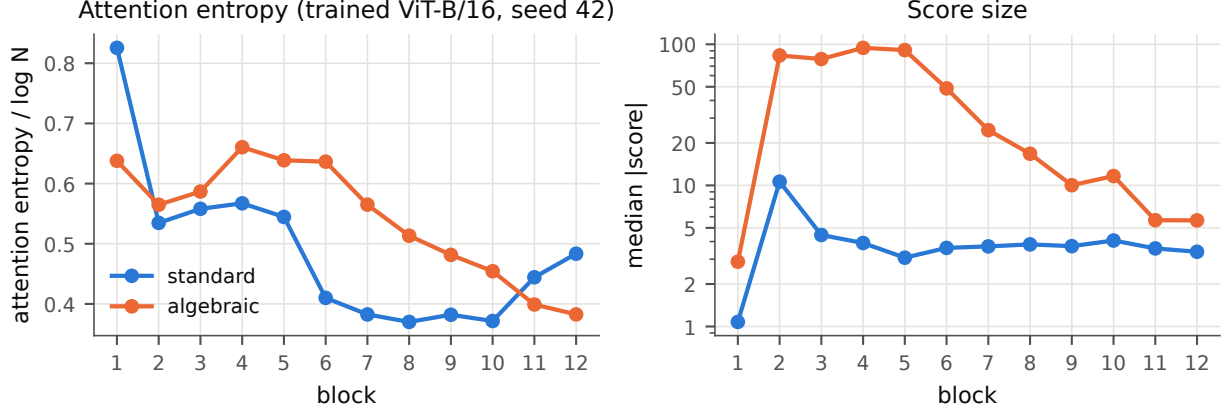


Figure 5: Attention in the trained ViT-B/16 models (seed 42, 256 validation images): mean entropy of the attention rows, normalised by $\log N$ (left), and median absolute score $|q \cdot k|/\sqrt{d_h}$ (right), per block.

Table 3: Finite precision: float32 error of each primitive and the perturbation caused by rounding its input to bfloat16, at input scale 30 (attention: total variation of weights; loss: relative error of the logit gradient). Relative to float64 [references](#).

Primitive	float32 error	bf16-input effect
softmax attention	1.5×10^{-6}	1.2×10^{-1}
radical attention (E_8)	3.0×10^{-7}	1.0×10^{-2}
cross-entropy (gradient)	3.2×10^{-7}	1.9×10^{-2}
power score $a = 63/64$ (gradient)	3.5×10^{-7}	1.0×10^{-2}

7 Verification

`analysis/symbolic_checks.py` checks every identity, derivative and bound used here with `sympy` and 50-digit `mpmath` (50 checks). The Lean 4 library `formal/AlgebraicVision` [8, 21] proves the statements cited in section 4 over the reals, using only the standard axioms. The proofs do not cover floating-point behaviour, which is measured instead (table 3), or optimisation and generalisation.

8 Limitations

The budgets are short (at most about 10 epochs) and the data is a single dataset at two resolutions; whether the gap persists with longer schedules, larger models, stronger augmentation and regularisation, or on transfer tasks is open. Three seeds per model suffice here only because the paired differences are consistent. Much of the advantage comes with robustness to the learning rate: the standard ViT has a narrow optimum, and stabilisers such as QK normalisation [9] or other schedules may narrow the gap; we did not test them. Relative positions explain most of the gain over the standard ViT; against the RoPE-2D control, which we did not tune separately, the lead is 2.31 points (table 2). The algebraic model is currently slower per step ($0.87\times$ of the standard model’s throughput).

9 Conclusion

An all-algebraic Vision Transformer, built from a polynomial-tailed radical attention kernel, a radical-logistic gate, rational rotations and a strictly proper power score, trains more stably and reaches higher ImageNet accuracy than the standard ViT at matched data, parameters, tokens and tuning, from small trials to ViT-B/16, where it leads by +9.86 points after 2.5B patch tokens. In this setting the algebraic choices are

improvements, not just substitutes.

Code and data. The code, the record of every run, the analysis scripts and the Lean proofs are at <https://github.com/tasmaikeni13/algebraic-vision>.

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